WEEK 4 · LESSON 20 OF 24 · HANDOUT

PCA: Seeing Many Features in Two

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=FgakZw6K1QQ — open the lesson online to watch inside the course.

🎓 **Go deeper:** StatQuest: PCA in Python — https://www.youtube.com/watch?v=Lsue2gEM9D0

🎓 **Go deeper:** freeCodeCamp: Linear Algebra for Machine Learning — full course — https://www.youtube.com/watch?v=rSjt1E9WHaQ

Your goal in this lesson: Explain what principal component analysis does, read explained variance, plot high-dimensional data in 2-D, and use PCA inside a model pipeline.

Too many features to look at

The wine dataset in scikit-learn has **13** chemical measurements for 178 wines from three grape varieties (cultivars). You cannot draw a 13-dimensional chart. **Principal component analysis (PCA)** finds new axes — **principal components** — that keep as much of the spread (variance) as possible in as few dimensions as possible.

- **PC1** is the direction in which the data varies most.
- **PC2** is the next most varied direction, at a right angle to PC1.
- And so on. Each component is a **weighted mix** of the original features.

It is like taking the best photo of a 3-D object: you choose the angle that shows the most detail.

PCA on the wine data

PCA is driven by variance, so features must be **standardised** first — otherwise the feature with the biggest units (proline, in the hundreds) would dominate.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

wine = load_wine(as_frame=True)
X_wine = StandardScaler().fit_transform(wine.data)

pca = PCA().fit(X_wine)
print(pca.explained_variance_ratio_.round(3))
print(pca.explained_variance_ratio_.cumsum().round(3))
```

- PC1 explains about **36%** of the variance and PC2 about **19%**: together about **55%** with only two numbers per wine instead of 13.
- About 10 components are needed to keep 95% of the variance.

Plot 13 features in 2-D

```
pca2 = PCA(n_components=2)
coords = pca2.fit_transform(X_wine)
plt.scatter(coords[:, 0], coords[:, 1], c=wine.target, s=15)
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("178 wines, 13 measurements, shown in 2-D")
plt.show()
```

The three varieties form three clearly separated clouds — even though PCA **never saw the labels**. That shows the 13 measurements really contain the information needed to tell them apart.

What does a component mean?

```
loadings = pd.Series(pca2.components_[0], index=wine.data.columns).sort_values()
print(loadings.round(2))
```

PC1 has large positive weights for flavanoids and total phenols and negative weights for non-flavanoid phenols and malic acid — so PC1 is roughly a "phenol richness" axis. Components are mixtures, so their meaning is often fuzzy. That is the price of PCA.

PCA before a model

PCA can make data smaller and remove noise before a model. Let's try it on the breast-cancer data (30 features):

```
from sklearn.datasets import load_breast_cancer
from sklearn.pipeline import make_pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import cross_val_score

cancer = load_breast_cancer()
for n in [2, 5, 10]:
    pipe = make_pipeline(StandardScaler(), PCA(n_components=n),
LogisticRegression(max_iter=1000))
    print(n, "components:", cross_val_score(pipe, cancer.data, cancer.target,
cv=5).mean().round(3))
full = make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000))
print("all 30 features:", cross_val_score(full, cancer.data, cancer.target,
cv=5).mean().round(3))
```

With just **2** components the model already scores about **0.95**; with 10 it matches all 30 features (about **0.98**). You can also ask for a share of variance: `PCA(n_components=0.95)` keeps as many components as needed for 95%.

When to use PCA

Good for	Be careful
Visualising many features	Components are hard to explain
Compressing data, speeding up models	Throws away some information
Removing correlated, noisy features	Needs scaling first
Image and gene data with thousands of features	Tree models rarely need it

For pictures of complex data, **t-SNE** and **UMAP** often show clusters more clearly, but they are only for visualisation.

Key terms

- **Dimensionality reduction** — representing data with fewer features.
- **Principal component** — a new axis that captures as much variance as possible.
- **Explained variance ratio** — the share of the total variance kept by each component.
- **Loadings** — the weights of the original features in a component.

Remember this: PCA rotates the data so the first few components keep most of the information; it is excellent for 2-D pictures and compression, at the cost of easy interpretation.

Name: _____ Date: _____

Activity

Task 1

Plot the cumulative explained variance of the wine data (`plt.plot(...cumsum())`) and mark the 90% line.

Task 2

Run PCA on the wine data **without** scaling. How much does PC1 explain now, and which feature dominates it? Why?

Task 3

Use `PCA(n_components=0.95)` in the cancer pipeline. How many components does it keep (`pipe[1].n_components_`)?

Task 4

Run PCA with 2 components on the customer data from Lesson 19 and colour the plot by k-means segment.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What does PCA do?
   - A. Removes rows
   - B. Adds labels
   - C. Predicts rent
   - D. Finds new axes that keep the most variance in fewer dimensions
2. Why must features be standardised before PCA?
   - A. PCA needs text
   - B. It is not needed
   - C. Otherwise features with big units dominate the components
   - D. To add noise
3. PC1 explains 36% and PC2 19% of the variance. How much do they explain together?
   - A. 17%
   - B. 55%
   - C. 36%
   - D. 100%
4. Did PCA use the wine labels to separate the varieties?
   - A. Only after scaling
   - B. Only for PC2
   - C. Yes
   - D. No — it is unsupervised; the separation comes from the measurements
5. What is the main cost of using PCA?
   - A. It always lowers accuracy to 50%
   - B. Components are mixtures that are hard to explain
   - C. It needs labels
   - D. It only works on images
6. What does PCA(n_components=0.95) do?
   - A. Keeps 95 components
   - B. Sets a threshold of 0.95
   - C. Keeps as many components as needed to explain 95% of the variance
   - D. Removes 95% of rows

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